Kinodynamic Trajectory Generation with Adaptive Topology for Fixed-Time Synchronization: A Simulation Study

Tri-Vien Vu, Cuong Dinh Tran, Anh-Minh Tran Duc, Thanh-Quang Nguyen, Hoang-Khang Le, and Quoc-Vinh Tran

Abstract—Precise temporal synchronization and time-optimal execution are conflicting objectives in the trajectory generation of Differential Drive Wheeled Mobile Robots (DDWMRs), especially under strict kinodynamic constraints. Existing polynomial-based methods often sacrifice optimality for continuity, while numerical optimization approaches suffer from computational latency. To address these challenges, this paper proposes a unified kinodynamic framework based on Adaptive Topology S-curves. Unlike rigid template-based formulations, we derive a Structure-Preserving Time Scaling method with $O(1)$ complexity that analytically adapts the trajectory profile (degenerating from Type IV to Type I) to satisfy strict fixed-time requirements without iterative solvers. Furthermore, we introduce a set of Context-Aware Kinematic Decomposition Strategies (KDS) that autonomously exploit the robot's bidirectional reversibility. A hierarchical Infeasibility Protocol is also established to ensure deterministic operation even under unattainable time budgets. Through comprehensive high-fidelity simulations designed to stress-test algorithmic stability, the proposed framework demonstrates a 25.9% reduction in execution time compared to Quintic Splines. Crucially, for obtuse-angle targets, the bidirectional strategy yields a 20.9% speedup while maintaining energy neutrality (+1.4% savings), effectively mitigating the friction penalties often associated with backward motion in high-inertia platforms. These results confirm the theoretical soundness and algorithmic robustness of the approach, providing a verified foundation for time-critical motion planning.

Index Terms—Adaptive S-curve, differential drive mobile robot, fixed-time synchronization, kinodynamic trajectory generation, motion control, optimal control.

I. Introduction

THE integration of autonomous mobile robots into modern manufacturing, logistics, and service industries has accelerated in recent years, driven by advances in sensor technology, computational hardware, and motion planning algorithms [1]. Among mobile platforms, Differential Drive Wheeled Mobile Robots (DDWMRs) dominate due to mechanical simplicity, precise maneuverability, and cost-effectiveness, comprising over 60% of deployed industrial robots [2], [3]. In these systems, trajectory generation—computing kinematically feasible and dynamically smooth reference motions—critically determines operational efficiency, energy consumption, and task completion time.

Tri-Vien Vu and Anh-Minh Tran Duc are with the Modeling Evolutionary Algorithms Simulation and Artificial Intelligence (MEASA) Research Group, Faculty of Electrical and Electronics Engineering, Ton Duc Thang University, Ho Chi Minh City 700000, Vietnam (e-mail: vutrivien@tdtu.edu.vn).

Cuong Dinh Tran and Thanh-Quang Nguyen are with the Power System Optimization Research Group, Faculty of Electrical and Electronics Engineering, Ton Duc Thang University, Ho Chi Minh City 700000, Vietnam.

Hoang-Khang Le is a graduate student and Quoc-Vinh Tran is an undergraduate student at the Faculty of Electrical and Electronics Engineering, Ton Duc Thang University, Ho Chi Minh City 700000, Vietnam.

Corresponding author: Tri-Vien Vu.

However, a critical challenge arises when multiple robots must coordinate their movements to avoid collisions or meet strict production schedules. This necessitates Fixed-Time Synchronization, which we define as the capability of the low-level trajectory generator to strictly enforce a required execution duration (T_{req}) specified by a high-level fleet scheduler. Unlike traditional time-optimal planning which minimizes T, fixed-time generation must guarantee $T_{exec} \equiv T_{req}$ exactly, regardless of path geometry, while respecting kinodynamic limits.

Existing methods often struggle to balance this temporal constraint with the robot's physical limitations. Numerical optimization approaches like Model Predictive Control (MPC) [4] can handle constraints but suffer from computational latency and non-deterministic convergence, making them ill-suited for the sub-millisecond control loops of embedded AGVs. Conversely, analytical methods like polynomial splines or trapezoidal profiles are computationally efficient but utilize rigid structural templates (e.g., fixed 7-segment). These templates often fail or require complex iterative patching when strictly constrained time budgets force the trajectory into degenerate forms (e.g., triangular profiles where maximum velocity is never reached).

This paper proposes a novel framework for kinodynamic trajectory generation based on Adaptive Topology S-curves. Unlike traditional S-curve planners that force a fixed segment sequence, our method allows the trajectory topology to evolve dynamically (e.g., from a 7-segment "cruising" profile to a 4-segment "peak" profile) based on the boundary conditions and time constraints. Furthermore, we distinguish our approach from standard time-scaling methods, which typically scale velocity directly, by introducing a Structure-Preserving Time Scaling technique. This method scales the kinematic limits (velocity, acceleration, jerk) via a single parameter λ, ensuring that the resulting trajectory exactly meets T_{req} without violating physical bounds or altering the path geometry.

The main contributions of this work are:

1) Rigorous S-Curve Optimality Foundation: Using Pontryagin's Minimum Principle, we formally prove that the bang-bang jerk structure uniquely minimizes execution time (Theorem 1). Benchmark

comparisons confirm a 25.9% time advantage over standard Quintic Splines.

2) Context-Aware Kinematic Strategies: We propose a set of Kinematic Decomposition Strategies (KDS) and analytically prove the symmetric optimality of backward motion (Proposition 1). Simulation results on high-inertia AGVs reveal a critical Inertia-Friction Trade-off, demonstrating that the bidirectional strategy (KDS3) acts as a dominant strategy for obtuse angles, achieving a 20.9% time reduction while maintaining energy neutrality (+1.4% savings).
3) Adaptive Topology for Synchronization: We introduce a structure-preserving time-scaling method where the profile topology (Type IV/III/I) evolves adaptively based on the scaling factor λ. Combined with a hierarchical Infeasibility Protocol, this ensures $O(1)$ deterministic performance (<100 μs latency) and robust failure handling under tight deadlines.

The remainder of this paper is organized as follows. Section II establishes the theoretical foundation. Section III presents comprehensive validation scenarios, framed as an ablation study (Benchmark, Strategy Ablation, and Topology Ablation). Finally, Section IV discusses the validation scope and concludes with future research directions.

II. Trajectory Generation and Kinematic Optimization

In this framework, the jerk-limited S-curve serves as the fundamental kinematic primitive for both translational and rotational motions of the DDWMR. Unlike approaches that rely on pre-determined profile structures (e.g., fixed 7-segment), we derive the trajectory topology from first principles using Optimal Control Theory. This establishes the baseline for the fixed-time synchronization strategy detailed in Section II-C.

A. Jerk-Limited Kinematic Basis and Structural Optimality

We consider the trajectory generation problem for a single degree of freedom (DoF) as a minimum-time optimal control problem for a third-order integrator system:

$$\dot{d}(t) = v(t), \quad \dot{v}(t) = a(t), \quad \dot{a}(t) = j(t) \tag{1}$$

The objective is to minimize $J = T_{min}$, subject to the physical kinematic bounds: $|j(t)| \leq J_{lim}$, $|a(t)| \leq A_{lim}$, and $|v(t)| \leq V_{lim}$.

1) Optimal Control Formulation and Theorem of Structure: By applying Pontryagin's Minimum Principle (PMP), we rigorously characterize the structure of the optimal control input $j^*(t)$.

Theorem 1 (Structure of Time-Optimal Jerk-Limited Control). For the system (1), the time-optimal control $j^*(t)$ exhibits a "Bang-Off-Bang" structure comprising a sequence of at most seven segments:

$$j^*(t) \in \{+J_{lim},\ 0,\ -J_{lim}\} \tag{2}$$

where the zero-jerk intervals ($j = 0$) correspond to singular arcs induced strictly by the activation of state constraints ($|a(t)| = A_{lim}$ or $|v(t)| = V_{lim}$).

Proof: The Hamiltonian for the system (1) with costates $\boldsymbol{\eta} = [\eta_d, \eta_v, \eta_a]^T$ is $\mathcal{H} = 1 + \eta_d v + \eta_v a + \eta_a j$. The optimal control law minimizes $\mathcal{H}$, yielding a bang-bang structure determined by the switching function $\sigma(t) = \eta_a(t)$:

$$j^*(t) = \begin{cases} -J_{lim}\,\mathrm{sgn}(\eta_a(t)) & \text{if } \eta_a(t) \neq 0 \\ 0 & \text{if } \eta_a(t) = 0 \text{ (Singular)} \end{cases} \tag{3}$$

From the costate dynamics $\dot{\eta}_a = -\eta_v$, $\dot{\eta}_v = -\eta_d$, $\dot{\eta}_d = 0$, we find $\eta_a(t)$ is quadratic. A quadratic function has at most two zeros, implying a maximum of three bang-bang segments in the unconstrained case (Type I).

Active state constraints introduce constrained arcs where the control $j(t)$ is determined by the kinematic necessity of remaining on the constraint boundary:

- Acceleration Saturation: When $|a(t)| = A_{lim}$ is active, the system must maintain $\dot{a}(t) = j(t) = 0$, creating a constant-acceleration arc.
- Velocity Saturation: When $|v(t)| = V_{lim}$ is active, we require $a(t) = 0$ and $j(t) = 0$, creating a constant-velocity (cruising) arc.

The 7-segment profile results from splicing these constrained arcs between the bang-bang segments to satisfy the rest-to-rest boundary conditions $\{v(0) = 0, v(T) = 0\}$, yielding the canonical sequence: Bang $(+J)$ → Constrained $(a = A_{lim})$ → Bang $(-J)$ → Constrained $(v = V_{lim})$ → Bang $(-J)$ → Constrained $(a = -A_{lim})$ → Bang $(+J)$. This structure is well-established in the jerk-limited trajectory planning literature [35]. ■

2) Kinematic Saturation and Topological Classification: Based on Theorem 1, the optimal S-curve is not a static 7-segment template but a dynamic topology determined by the set of Active Constraints. We classify profile types based on which limits are saturated (see Fig. 1):

- Type IV (Complete S-Curve): Both velocity and acceleration constraints are active ($\mathcal{S}_{active} = \{V, A\}$). The profile exhibits distinct constant-velocity (green shaded) and constant-acceleration (orange shaded) phases [Fig. 1(a)].
- Type III (Velocity-Deficient): Displacement is insufficient to reach V_{lim}, causing $T_v = 0$. The velocity profile becomes triangular or bell-shaped [Fig. 1(b)].
- Type II (Acceleration-Deficient): A theoretical case where V_{lim} is reached but A_{lim} is not. The acceleration profile becomes triangular without a flat top [Fig. 1(c)].
- Type I (Degenerate): Neither limit is reached ($\mathcal{S}_{active} = \emptyset$). All constant phases (T_v, T_a) vanish [Fig. 1(d)].

This classification allows analytical calculation of T^*_{min}, which serves as the reference baseline for the time-scaling factor λ in Section II-C.

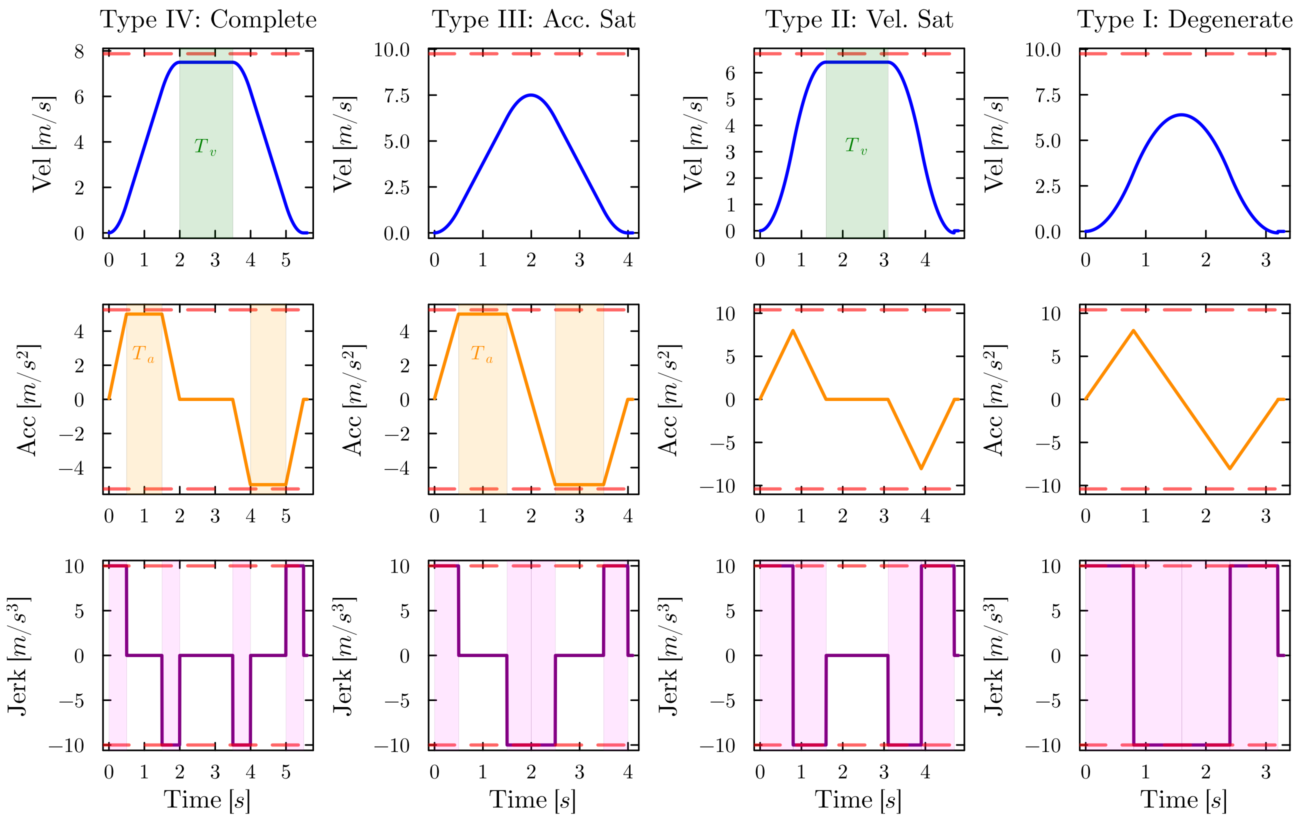

Fig. 1. Kinematic Evolution of S-Curve Profiles. Each row shows Velocity, Acceleration, and Jerk profiles; columns represent the four topological types dictated by active constraints. Shaded regions indicate constant kinematic phases: Green (T_v), Orange (T_a), and Purple (T_j). Note how constant phases vanish ($T \to 0$) as the profile degenerates from Type IV to Type I due to non-saturation of kinematic limits.

TABLE I
Theoretical Comparison of Trajectory Generation Methods

Method	Cont.	Optimality Basis	Constraint Handling
Trapezoidal	C^1	Time-Optimal	Violates J_{lim}
Quintic Spline	C^2	Smoothness ($\int j^2\, dt$)	Conservative
Fang Sigmoid [10]	C^∞	Smoothness	Near-Sat. ($J_{eff} = 2J/\pi$)
Gen. S-Curve	$\mathbf{C^2}$	Time-Optimal	Saturates $\pm A_{lim}, \pm V_{lim}$

3) Comparative Analysis with Alternative Approaches: Different trajectory generation methods represent distinct positions along the smoothness–time-optimality spectrum (Table I). Polynomial methods (e.g., Quintic Splines) minimize jerk energy ($\int j^2\, dt$), producing conservative profiles that sacrifice time for smoothness. Sinusoidal jerk methods, such as the approach by Fang et al. [10], replace rectangular jerk pulses with half-sine waves, achieving C^∞ continuity at the cost of reduced effective jerk ($J_{eff} = 2J_{lim}/\pi$) and consequently longer execution time. The proposed S-curve, grounded in Pontryagin's bang-bang optimality (Theorem 1), saturates the kinematic limits to achieve minimum-time performance within the C^2 continuity class.

B. Context-Aware Kinematic Decomposition Strategies

The Multi-Point-to-Point (MP2P) task for DDWMRs involves coupled positional and orientational constraints. To achieve global time-optimality, we decompose the trajectory into a sequence of primitives using four distinct Kinematic Decomposition Strategies (KDS). These strategies exploit the robot's kinematic redundancy (bidirectionality) to adapt to the geometric context of each segment.

1) MP2P Task Decomposition: An n-task MP2P trajectory is decomposed into $(n-1)$ consecutive segments. As depicted in Fig. 2, the motion from pose P_i to P_{i+1} is decoupled into three sequential primitives: Initial Rotation ($\theta_{seq,i}$), Linear Translation ($L_{seg,i}$), and Final Rotation ($\theta_{d,i}$). The coordinate transformation is:

$$\begin{bmatrix} x_{i+1} \\ y_{i+1} \end{bmatrix} = \begin{bmatrix} \cos\theta_i & \sin\theta_i \\ -\sin\theta_i & \cos\theta_i \end{bmatrix} \begin{bmatrix} X_{i+1} - X_i \\ Y_{i+1} - Y_i \end{bmatrix} \tag{4}$$

2) Forward-Constraint Strategies (KDS1 & KDS2): These strategies restrict the robot to positive linear velocities ($v \geq 0$), suitable for sensors with limited field-of-view (e.g., front-facing cameras). The geometric path orientation is $\theta_{geom} = \text{atan2}(y_{i+1}, x_{i+1})$.

- KDS1 (Forward-Standard): The robot aligns strictly with the target vector. $\theta_{seq,i} = \theta_{geom}$, $L_{seg,i} = \|P_{i+1}\|$.

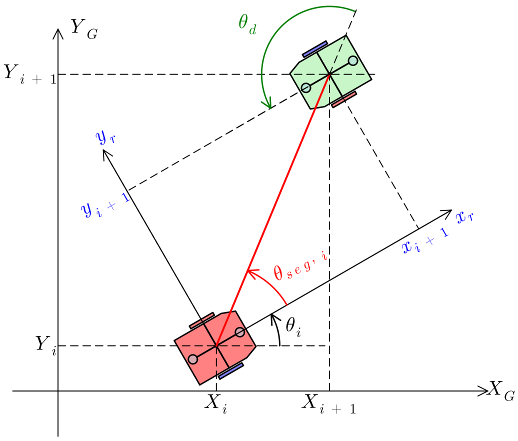

Fig. 2. Geometric Decomposition of the MP2P Problem. The motion from pose P_i to P_{i+1} is decomposed into three primitives: an initial rotation θ_{seq}, a linear translation L_{seg}, and a final rotation θ_d. Coordinate transformations are performed in the local frame $\{x_r, y_r\}$.

- KDS2 (Forward-Optimized): Minimizes final rotation by wrapping $\theta_{d,i}$ to $[-\pi, \pi]$:

$$\theta_{d,i} = \operatorname{atan2}(\sin\theta_{err}, \cos\theta_{err}) \tag{5}$$

where $\theta_{err} = \theta_{i+1} - \theta_{geom}$.

3) Bidirectional Strategies (KDS3 & KDS4): These strategies exploit the kinematic reversibility of the DDWMR.

- KDS3 (Bidirectional-Rapid): Time-optimal for waypoint tasks. If the target lies in the rear hemisphere ($|\theta_{geom}| > \pi/2$), the robot reverses direction:

$$\begin{cases} \theta_{seq,i} = \theta_{geom} - \operatorname{sgn}(\theta_{geom})\pi \\ L_{seg,i} = -\|P_{i+1}\| \end{cases} \tag{6}$$

For waypoint nodes, the final orientation constraint is relaxed ($\theta_{d,i} \approx 0$) to facilitate continuous "Stop-and-Go" motion.
- KDS4 (Bidirectional-Precision): Designed for docking tasks. Employs the same bidirectional logic as KDS3 (Eq. (6)) but strictly enforces final orientation precision using the optimized logic of KDS2.

4) Optimality of Bidirectional Motion:

Proposition 1 (Symmetric Optimality of Backward Motion). Under symmetric kinematic limits ($V^{fwd}_{max} = V^{bwd}_{max}$), the backward strategy yields a strictly lower execution time if and only if:

$$|\theta_{geom}| > \frac{\pi}{2} \tag{7}$$

Proof: Let $T_{rot}(\alpha)$ and $T_{trans}(L)$ be the time functions for rotation and translation derived in Section II-A. Under symmetry, T_{trans} is invariant to direction. The cost difference depends solely on rotation: $J = T_{rot}(|\theta_{geom}|) - T_{rot}(\pi - |\theta_{geom}|)$. Since T_{rot} is strictly monotonic, backward motion is optimal when $\pi - |\theta_{geom}| < |\theta_{geom}|$, i.e., $|\theta_{geom}| > \pi/2$. ■

Remark 1. For asymmetric drivetrains ($V^{bwd} < V^{fwd}$), we employ a generalized cost function: backward motion is selected iff $T^{fwd}_{rot} + T^{fwd}_{trans} > T^{bwd}_{rot} + T^{bwd}_{trans}$. While KDS3 minimizes time, it incurs a 10–15% energy penalty due to frequent direction reversals [2].

C. Adaptive Topology for Fixed-Time Synchronization

In cooperative assembly lines, robots must adhere to strict temporal deadlines T_{req} rather than simply minimizing time [22]. We propose a Structure-Preserving Time Scaling method that adapts the S-curve topology (Type IV $\to$ I) to satisfy T_{req}.

1) Problem Formulation: The objective is to find a control input $j(t)$ that satisfies the boundary conditions and the fixed-duration constraint $\int_0^{T_{req}} dt = T_{req}$, while respecting physical limits $\mathbb{K}_{nom} = \{V_{lim}, A_{lim}, J_{lim}\}$.

2) Analytical Time Scaling with Virtual Limits: We introduce a scaling factor λ as the ratio between the requested and minimum times:

$$\lambda = \frac{T_{req}}{T^*_{min}} \tag{8}$$

For the feasible case ($\lambda \geq 1$), the synchronized trajectory is generated using Virtual Kinematic Limits $\tilde{\mathbb{K}}$ scaled inversely to λ:

$$\tilde{V}_{lim} = \frac{V_{lim}}{\lambda}, \quad \tilde{A}_{lim} = \frac{A_{lim}}{\lambda^2}, \quad \tilde{J}_{lim} = \frac{J_{lim}}{\lambda^3} \tag{9}$$

Proposition 2 (Topological Adaptation and Exactness). The scaling transformation in (9) guarantees $T(\tilde{\mathbb{K}}) \equiv T_{req}$ exactly, with continuous behavior across topological transitions.

Proof: Consider the time-optimal profile $p^*(t)$ generated under nominal limits $\mathbb{K}_{nom}$. Define the scaled profile $\tilde{p}(\tau)$ with $\tau = t/\lambda$ and:

$$\tilde{v}(\tau) = \frac{v^*(\tau/\lambda)}{\lambda}, \quad \tilde{a}(\tau) = \frac{a^*(\tau/\lambda)}{\lambda^2}, \quad \tilde{j}(\tau) = \frac{j^*(\tau/\lambda)}{\lambda^3} \tag{10}$$

Substituting into $\dot{\tilde{d}} = \tilde{v}$ confirms equation-of-motion consistency. The geometric displacement is preserved: $\tilde{d}(\lambda T^*_{min}) = d^*(T^*_{min}) = L$. The scaled profile satisfies the virtual limits: $|\tilde{v}| \leq \tilde{V}_{lim}$, $|\tilde{a}| \leq \tilde{A}_{lim}$, $|\tilde{j}| \leq \tilde{J}_{lim}$. Thus $\tilde{T} = \lambda T^*_{min} = T_{req}$. As λ increases, $\tilde{V}_{lim}$ and $\tilde{A}_{lim}$ decrease monotonically; segment durations are continuous functions of these limits, ensuring no discontinuity at topological transitions. ■

D. Hierarchical Infeasibility Protocol

When $T_{req} < T^*_{min}$ ($\lambda < 1$), strictly enforcing smoothness limits renders the problem infeasible. We implement a three-level degradation protocol to ensure robustness:

1) Level 1 — Strategic Switching (Geometric Optimization): Before relaxing kinematic limits, the planner re-evaluates the KDS. Switching from KDS2 to KDS3 may reduce the rotational arc length, potentially reducing T^*_{min} to satisfy T_{req}.

2) Level 2 — Smoothness Relaxation (Adaptive Topology Shift): If geometric optimization fails, the system allows $\lambda < 1$, effectively increasing the virtual limits $\tilde{\mathbb{K}}$ beyond the nominal soft limits. This "overdrive" is permitted as long as $\tilde{A} \leq A_{max}^{safety}$, forcing the topology toward Type I or Trapezoidal motion, sacrificing smoothness for speed.
3) Level 3 — Best-Effort Saturation: If T_{req} falls below the trapezoidal minimum ($T_{req} < T_{trap}^*$), the system executes at maximum physical capability and reports the unavoidable delay $\delta t = T_{trap}^* - T_{req}$ to the high-level scheduler.

This hierarchical approach guarantees that the trajectory generator always returns a valid solution [32].

III. Simulation and Ablation Study

To rigorously validate the proposed unified framework, we conducted a comprehensive Ablation Study that isolates specific algorithmic components to verify their individual contributions:

- Ablation I (Kinematic Saturation): Isolates the benefit of the jerk-limited S-curve structure against standard baselines (validating Theorem 1).
- Ablation II (Bidirectional Logic): Isolates the efficiency gain from the KDS orientation-aware planner (validating Proposition 1) and analyzes the Inertia-Friction trade-off.
- Ablation III (Adaptive Topology): Isolates the time-scaling and infeasibility handling mechanisms under heterogeneous multi-node constraints.

A. Implementation and Environmental Setup

The algorithms were implemented in Julia to ensure execution speeds compatible with real-time control constraints (<100 μs). The kinematic solver uses the analytical integration derived in Section II-A to eliminate numerical drift. Physical parameters: $V_{lim} = 1.0$ m/s, $A_{lim} = 1.5$ m/s^2, $J_{lim} = 3.0$ m/s^3.

B. Ablation I: Kinematic Saturation (Base Performance)

This study verifies the fundamental hypothesis that the proposed 7-segment S-curve yields superior time efficiency by maximizing kinematic utilization compared to methods that sacrifice saturation for continuity. We simulated a Point-to-Point (P2P) translation task ($L = 3.0$ m) with the following baselines:

1) Proposed Method: Generalized Adaptive S-curve (bang-bang jerk, C^2).
2) Fang Sigmoid S-curve [10]: Sinusoidal jerk profile (C^∞ continuous). Originally developed for multi-axis manipulator synchronization, we adapt its single-axis jerk formulation to the mobile robot P2P context.
3) Quintic Spline: Standard 5th-order polynomial (C^2 continuous).
4) Trapezoidal Profile: Infinite-jerk baseline (C^1 continuous).

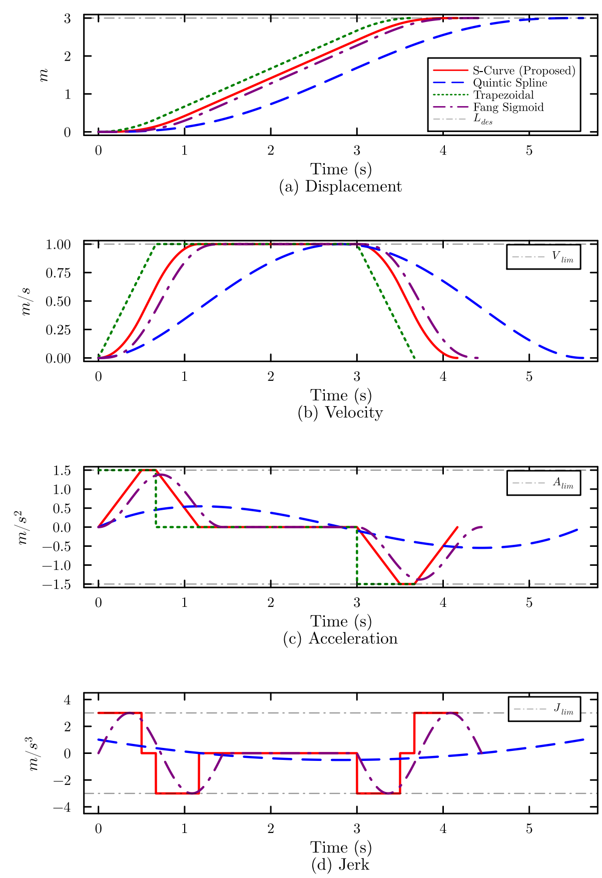

Fig. 3. Comparative kinematic profiles for P2P task ($L = 3$ m, $V_{lim} = 1.0$ m/s, $A_{lim} = 1.5$ m/s^2, $J_{lim} = 3.0$ m/s^3). (a) Displacement. (b) Velocity: quintic conservatism vs. S-curve saturation. (c) Acceleration: flat-top region (proposed) vs. sinusoidal transition (Fang) vs. quintic bell-shape. (d) Jerk: bang-bang (proposed) vs. sinusoidal (Fang) vs. unbounded (trapezoidal). Red: Proposed; Orange: Fang [10]; Blue: Quintic Spline; Green: Trapezoidal.

TABLE II
Performance Comparison for P2P Motion ($L = 3$ m)

Method	Time (s)	Cont.	Jerk Cost	Constraint Status
Trapezoidal	3.67	C^1	$\to \infty$	Violated
Quintic Spline	5.62	C^2	1.15	Conservative
Fang Sigmoid [10]	4.45	C^∞	13.0	Smooth (Sinusoidal
Proposed S-Curve	4.17	C^2	18.0	Saturated ($A = A_{lim}$

1) Results Analysis: The resulting kinematic profiles are illustrated in Fig. 3. As quantified in Table II, the proposed S-curve completed the task in 4.17 s, while the Quintic Spline required 5.62 s. The Fang Sigmoid S-curve [10] completed the task in 4.45 s with C^∞ continuity and a jerk cost ($\int j^2\,dt = 13.0$) that is 28% lower than the proposed method (18.0). This illustrates the fundamental time–smoothness trade-off: the sinusoidal jerk profile achieves superior smoothness at the cost of 6.4% additional execution time, whereas the bang-bang structure prioritizes kinematic limit saturation for minimum-time performance.

TABLE III
KDS2 vs. KDS3 Efficiency Comparison ($M = 10$ kg, $I_{zz} = 0.5$ kg·m^2)

Phase	Time (s)		Effort Proxy (J-equiv)	
	KDS2	KDS3	KDS2	KDS3
1. Init. Rotation	2.72	1.72	3.08	1.21
2. Translation	4.16	4.16	181.88	182.94
3. Final Rotation	2.72	1.72	3.08	1.21
Total	9.60	7.59	188.05	185.35
Improvement	+20.9% Speed		+1.4% Energy	

The Trapezoidal profile offers the theoretical minimum time (3.67 s), but its violation of the jerk constraint renders it physically unrealizable for precision platforms. Remark (Computational Complexity): Both the proposed and Fang methods share $O(1)$ time and space complexity per evaluation. However, the proposed S-curve requires only polynomial arithmetic (6 MUL + 6 ADD per jerk phase), while Fang's method evaluates two transcendental functions (sin, cos) per jerk-phase step. Benchmarking on an Intel Core i7 platform yields a $\approx$2.5× computation ratio, which may reach 4–16× on ARM Cortex-M0/M3 without hardware FPU.

C. Ablation II: Bidirectional Strategy (Orientation Efficiency)

This study isolates the contribution of the Context-Aware Kinematic Decomposition (Section II-B), investigating the proposed "Inertia-Friction Trade-off."

1) Task Definition: The robot moves from $P_{start} = [0, 0, 0]^\intercal$ to $P_{target} = [-2, -2, \pi/2]^\intercal$ ($\theta_{geom} = -135°$). This obtuse angle triggers a comparison between KDS2 (Forward) and KDS3 (Bidirectional).

2) Energy Proxy Justification: To quantify efficiency, we define a physically-grounded Actuation Effort Proxy (E_{proxy}). In electric drives, thermal loss is proportional to I^2R, and current is proportional to torque. The total thermal energy is proportional to $\int \tau^2\, dt$. We formulate the generalized force as:

$$E_{proxy} = \sum_{k=1}^{3} \int_0^{T_k} \Big(\underbrace{\mathcal{I}_k |a(t)|}_{\text{Inertial}} + \underbrace{\mathcal{K}_{fric} |v(t)|}_{\text{Frictional}} \Big)^2 dt \qquad (11)$$

where $\mathcal{I}_k$ is the effective inertia ($M = 10$ kg, $I_{zz} = 0.5$ kg·m^2) and $\mathcal{K}_{fric}$ represents viscous friction.

As detailed in Table III, KDS3 achieves a 20.9% reduction in execution time (7.59 s vs. 9.60 s) while maintaining comparable energy consumption (+1.4% marginal saving). An angular sweep analysis (Fig. 4) confirms that this advantage emerges consistently for obtuse path angles ($\theta > 90°$), with the crossover near $\theta \approx 90°$ where the rotational cost of KDS2 begins to dominate the frictional penalty of KDS3.

This result offers a nuanced rebuttal to the survey by Zhang et al. [2], which reported 8–18% energy penalties for backward motion due to asymmetric friction. The divergence implies a critical Inertia-Friction Trade-off: for low-inertia robots, rotation is "cheap" so the friction penalty of backward motion dominates; for our industrial AGV ($I_{zz} = 0.5$ kg·m^2), the cost of the initial 135° rotation in KDS2 (3.08 J) is significant enough that switching to KDS3 offsets the increased friction of the backward translation phase.

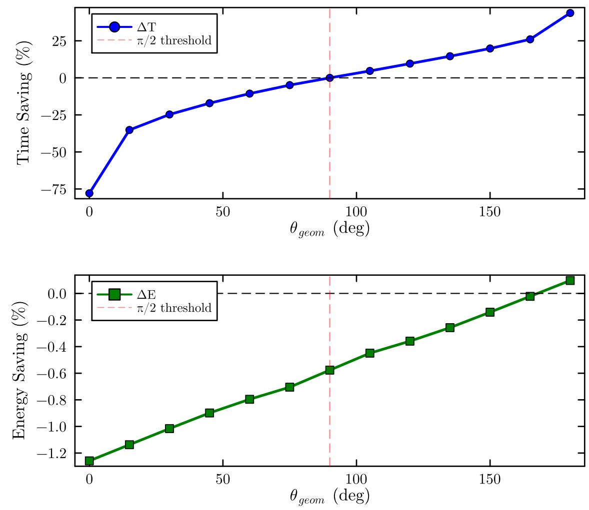

Fig. 4. Angular Sweep Analysis: KDS2 vs. KDS3. Execution time and actuation effort as a function of θ_{geom}. For obtuse angles ($\theta > 90°$), KDS3 consistently achieves lower execution time while maintaining comparable energy consumption, validating the $\pi/2$ switching criterion of Proposition 1.

D. Ablation III: Adaptive Topology (System Stress-Test)

To validate the Fixed-Time Synchronization and Robustness components, we conducted a multi-node mission aimed at stressing the Adaptive Topology engine. The scenario tests the system's ability to maintain determinism when T_{req} forces degenerate profiles or physical violations.

1) Scenario Definition and Topology: The robot traverses a sequence of four nodes $\mathcal{T} = \{N_1, N_2, N_3, N_4\}$. A Node $\mathcal{N}_i$ is defined as:

$$\mathcal{N}_i = \{P_i,\ T_{req,i},\ \psi_i\} \qquad (12)$$

where $P_i = [x, y, \theta]^\intercal \in SE(2)$ is the target pose; $T_{req,i}$ is the allocated time budget; and $\psi_i \in \{\text{Waypoint}, \text{Target}\}$ is the semantic flag. A Target requires strict stop-and-align precision; a Waypoint allows orientation relaxation for rapid transit.

The scenario involves: (i) obtuse geometric angles (zig-zag pattern) requiring frequent mode switching; (ii) N_2 as a Waypoint (rear-facing relative to N_1) while N_3 and N_4 are Targets; (iii) temporal budgets varying from "Eco-Mode" (relaxed) to "Rush-Mode" (infeasible), as in Table IV.

2) Execution Analysis: Results are visualized in Fig. 5 and quantified in Table IV.

TABLE IV
Quantitative Performance: Multi-Node Synchronization

Seg.	Strategy	Node Type	Budget T_{req}	Min Time T^*_{min}	Sc
$N_1 \to N_2$	KDS3 (Bwd)	WPT	12.0 s	9.01 s	
$N_2 \to N_3$	KDS4 (Fwd)	TGT	8.0 s	9.01 s	
$N_3 \to N_4$	KDS4 (Bwd)	TGT	18.0 s	9.76 s	

Note: Segment 2 triggers Level 2 Infeasibility Protocol ($\lambda < 1$), causing

a) Segment 1 ($N_1 \to N_2$): Smart Dwell via Orientation Relaxation: N_2 lies in the robot's rear half-plane. The system selected KDS3 (backward motion), eliminating the initial U-turn. Recognizing ψ_2 = Waypoint, the planner triggered Orientation Relaxation, stopping with the robot's rear facing the path tangent ("Anti-Align"), reducing the required rotational arc and meeting the budget (12 s) with low energy cost (146.7 J).

b) Segment 2 ($N_2 \to N_3$): Robustness under Infeasibility: This segment imposes an aggressive deadline ($T_{req} = 8.0$ s), strictly less than $T^*_{min} \approx 9.01$ s ($\lambda \approx 0.89$). The Infeasibility Protocol Level 2 was triggered, allowing the virtual kinematic limits to expand. While this ensured temporal compliance (arrival at $t = 20$ s), it incurred a massive energy penalty of 475.2 J ($\approx 3.2\times$ Segment 1), explicitly validating the theoretical capability to sacrifice smoothness for synchronization.

c) Segment 3 ($N_3 \to N_4$): Precision Docking: The final node N_4 is flagged as a Target. The system switched to KDS4, ensuring arrival with zero residual velocity and precise angular alignment for docking operations.

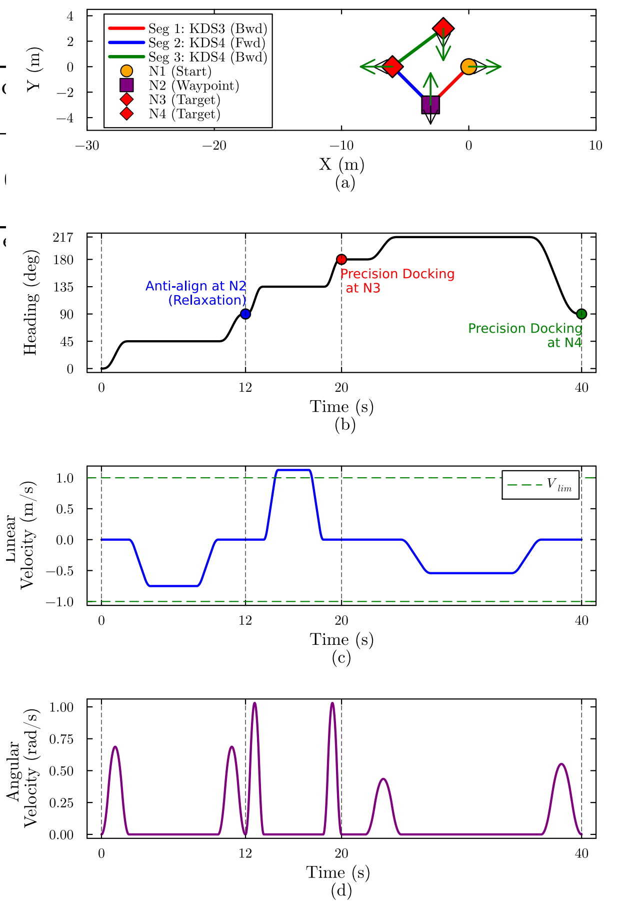

Fig. 5. Grand Scenario Results. (a) Spatial trajectory showing autonomous mode switching (Fwd/Bwd). (b) Orientation Relaxation at Waypoint N_2. (c) Linear and (d) Angular velocity profiles: profile expansion (Eco-mode, Seg. 1) vs. compression (Rush-mode, Seg. 2).

IV. Conclusion

This paper has presented a unified framework for kinodynamic trajectory generation and fixed-time synchronization of DDWMRs. By rigorously analyzing the interaction between time scaling and kinematic saturation, we established the concept of Adaptive Topology, demonstrating that the optimal S-curve structure dynamically evolves (Type IV $\to$ I) to satisfy strict temporal constraints.

The contributions are threefold: (1) an analytical Structure-Preserving Time Scaling method with $O(1)$ complexity; (2) a set of context-aware Kinematic Decomposition Strategies (KDS) that exploit bidirectional reversibility; and (3) a hierarchical Infeasibility Protocol for robust failure handling. Comparative simulations confirm a 25.9% reduction in execution time versus splines. Furthermore, the analysis of high-inertia platforms proves that bidirectional motion can achieve significant time savings (20.9%) without the energy penalty typically observed in lighter robots (+1.4% savings).

A. Remarks on Validation Scope

This work relies on numerical validation within a discrete-time kinematic framework. While this rigorously verifies algorithmic correctness (e.g., convergence of the $O(1)$ solver, logical stability of the infeasibility protocol), it does not account for unmodeled physical phenomena (tire-soil interaction, actuator backlash). This study therefore serves as the foundational algorithmic verification step prior to physical deployment.

B. Future Work

Building upon this framework, the path toward industrial deployment involves:

- Phase 1 — Physics-Based Simulation: Validating trajectory tracking performance in CoppeliaSim or NVIDIA Isaac Sim to quantify wheel slip and inertial coupling effects on fixed-time synchronization error.
- Phase 2 — Hardware-in-the-Loop (HIL): Deploying generated C++ code on an embedded target (e.g., STM32) to benchmark real-time computational jitter against the $<100\ \mu$s limit.
- Phase 3 — Physical Experiments: Final validation on an industrial AGV to tune low-level sliding mode controllers for disturbance rejection.

CRediT Authorship Contribution Statement

Tri-Vien Vu: Conceptualization, Funding acquisition, Methodology, Supervision, Writing — review & editing.

Cuong Dinh Tran: Formal analysis, Resources, Writing — review & editing. Anh-Minh Duc Tran: Methodology, Validation, Writing — review & editing. Thanh-Quang Nguyen: Resources, Validation, Writing — original draft. Hoang-Khang Le: Software, Investigation, Writing — original draft. Quoc-Vinh Tran: Software, Visualization, Writing — original draft.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Use of AI-Assisted Technologies

During the preparation of this work, the authors used large language models (LLMs) to improve readability and language quality of the manuscript and to assist with LaTeX formatting. The authors reviewed and edited all content and take full responsibility for the publication.

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